CIT
Technical University of Munich

Ecorrection
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Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Diskrete Wahrscheinlichkeitstheorie

Exam:	IN0018 / Retake	**Date:**	Friday 9th October, 2026
Examiner:	Niki Kilbertus	**Time:**	11:00 – 13:00

Sample Solution
Correction Notes

Working instructions

- This exam consists of **12 pages** with a total of **5 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 44 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **single sheet of A4 paper** with handwritten notes on both sides
  - one **analog dictionary** English ↔ native language **without notes**
- Subproblems marked by * can be solved without results of previous subproblems, using any facts supplied in the question. Read later parts even if you are stuck on an earlier one.
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- Place your student ID visibly on your table.

Left room from _____ to _____ / Early submission at _____

Problem 1 True or false (5 credits)

For each of the following statements, mark exactly one answer. No justification is required.

a) Let A and B be disjoint events with $P(A) > 0$ and $P(B) > 0$. Then A and B are independent.

☐ True

☒ False

False. Disjointness gives $P(A \cap B) = 0$, whereas $P(A)P(B) > 0$.

b) If the real-valued random variables X and Y are independent, then X^2 and Y^2 are independent as well.

☐ False

☒ True

True. Applying a measurable function separately to each of two independent random variables preserves independence.

c) Let X be a real-valued random variable with characteristic function ϕ_X. Then $\phi_{X+2}(t) = e^{2it}\phi_X(t)$ for all $t \in \mathbb{R}$.

☐ False

☒ True

True. $\mathbb{E}[e^{it(X+2)}] = e^{2it}\,\mathbb{E}[e^{itX}] = e^{2it}\phi_X(t)$.

d) Let $X_1, X_2, \ldots$ and X be real-valued random variables on a common probability space $(\Omega, \mathcal{A}, P)$. If $X_n \xrightarrow{a.s.} X$, then $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ for every $\omega \in \Omega$.

☒ False

☐ True

False. Almost sure convergence allows an exceptional set of probability zero. For example, take $\Omega = \{0, 1\}$, $P(\{0\}) = 1$, $X_n(\omega) = \omega$ for all n, and $X(\omega) = 0$.

e) A test at significance level $\alpha = 0.05$ has probability at most 0.05 of making a type I error under every parameter value in the null hypothesis.

☐ False

☒ True

True. By definition, a test at level α satisfies $\sup_{\theta\in\Theta_0} P_\theta(\text{reject } H_0) \le \alpha$. Under the null hypothesis, rejection is a type I error.

For each item: 1 point if exactly the correct answer is marked; 0 points otherwise, including both answers or neither answer. No justification is graded. The explanations above are for the sample solution only.

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Problem 2 True or false with justification (12 credits)

Decide whether each statement is true or false and provide a justification (proof, counterexample, or explanation via known results).

a) Let $\Omega = \{1,2,3,4,5,6\}$ and let

$$\mathcal{A} = \{\emptyset, \{1,2\}, \{3,4\}, \{5,6\}, \{1,2,3,4\}, \{1,2,5,6\}, \{3,4,5,6\}, \Omega\}\,.$$

Define $X : \Omega \to \mathbb{R}$ by $X(1) = X(2) = 0$, $X(3) = X(4) = X(5) = 1$, and $X(6) = 2$. Then X is measurable from $(\Omega, \mathcal{A})$ to $(\mathbb{R}, \mathcal{B})$, where $\mathcal{B}$ is the Borel σ-algebra.

0 1 2 3

False. The set $\{1\}$ is a Borel set, but

$$X^{-1}(\{1\}) = \{3,4,5\} \notin \mathcal{A}\,.$$

Thus the inverse image of a measurable set in the target need not belong to the source σ-algebra, so X is not measurable. Some other inverse images are measurable, for example $X^{-1}(\{0\}) = \{1,2\}$ and $X^{-1}(\{1,2\}) = \{3,4,5,6\}$, but this is insufficient.

1 point for false; 1 point for computing a suitable inverse image; 1 point for noting that the target set is Borel and its inverse image is not in $\mathcal{A}$, hence measurability fails. Accept, e.g., $X^{-1}(\{2\}) = \{6\}$ or $X^{-1}(\{0,1\}) = \{1,2,3,4,5\}$. The fact that the target set is Borel may be implicit for a singleton or interval. A correct explanation that X splits the atom $\{5,6\}$ also receives full justification credit. There is no need to enumerate all inverse images. Merely stating that $\mathcal{A}$ is not the power set is insufficient.

b) Let X and Y be independent real-valued random variables whose expectations and variances exist. If $\mathbb{E}[Y] = 0$, then $\operatorname{cov}[X, XY] = 0$.

0 1 2 3

True. By independence,

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] = 0, \qquad \mathbb{E}[X^2 Y] = \mathbb{E}[X^2]\,\mathbb{E}[Y] = 0\,.$$

It follows that

$$\operatorname{cov}[X, XY] = \mathbb{E}[X^2 Y] - \mathbb{E}[X]\,\mathbb{E}[XY] = 0\,.$$

The covariance is well-defined: independence and the finite second moments imply $\mathbb{E}[(XY)^2] = \mathbb{E}[X^2]\,\mathbb{E}[Y^2] < \infty$.

1 point for true; 1 point for the covariance expansion; 0.5 points each for correctly obtaining $\mathbb{E}[XY] = 0$ and $\mathbb{E}[X^2 Y] = 0$ using independence. Explicitly accept the step that X^2 and Y are independent because X and Y are independent; no proof of this preservation property is required. Also accept LOTUS applied to the factorized joint distribution (or density when applicable), a correct conditional-expectation argument, or the equivalent calculation $\operatorname{cov}[X, XY] = \operatorname{var}[X]\,\mathbb{E}[Y] = 0$. No separate integrability proof is required. Do not accept an unsupported claim that X and XY are independent.

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0 1 2 3

c) Let X and Y be real-valued random variables on a common probability space. If $\phi_X(t) = \phi_Y(t)$ for every $t \in \mathbb{R}$, then $P(X = Y) = 1$.

False. Let X be uniformly distributed on $\{-1, 1\}$ and define $Y := -X$. Both variables have the same distribution, and hence the same characteristic function. Explicitly,

$$\phi_X(t) = \phi_Y(t) = \frac{e^{it} + e^{-it}}{2} = \cos(t) .$$

However, $Y = -X \neq X$ for both possible values of X, so $P(X = Y) = 0$. Equality of characteristic functions determines equality of distributions, not equality of the random variables almost surely.
1 point for false; 1 point for a concrete example and a correct explanation that the characteristic functions agree; 1 point for showing that $P(X = Y) < 1$. No explicit characteristic-function computation is required if equality of distributions is established. For example, independent Ber$(1/2)$ variables also work, with $P(X = Y) = 1/2$. For an explanation via known results, award the 2 justification points for correctly identifying equality in distribution via characteristic-function uniqueness and invoking the known fact that equality in distribution does not imply almost sure equality.

0 1 2 3

d) Two tests of the same null hypothesis $H_0 : \theta \in \Theta_0$ against the same alternative are each at significance level 0.025. Both tests use the same data. The test that rejects H_0 whenever at least one of these two tests rejects is at significance level 0.05.

True. Let R_1 and R_2 be the rejection events of the two tests. For every $\theta \in \Theta_0$, $P_\theta(R_j) \leq 0.025$ for $j \in \{1, 2\}$. The combined rejection event is $R_1 \cup R_2$. By inclusion–exclusion,

$$\begin{aligned} P_\theta(R_1 \cup R_2) &= P_\theta(R_1) + P_\theta(R_2) - P_\theta(R_1 \cap R_2) \\ &\leq P_\theta(R_1) + P_\theta(R_2) \\ &\leq 0.025 + 0.025 = 0.05 . \end{aligned}$$

The first inequality holds because $P_\theta(R_1 \cap R_2) \geq 0$. Taking the supremum over $\theta \in \Theta_0$ proves that the combined test is at level 0.05.
1 point for true; 1 point for identifying the combined rejection event and the individual bounds under every null parameter; 1 point for the probability bound and conclusion of level 0.05. Accept inclusion–exclusion, subadditivity, a disjoint decomposition, or an equivalent argument with test indicators. An explicit supremum is not required if the argument applies to every null parameter. Multiplying acceptance probabilities without a justified independence assumption does not establish the claim. The actual size may be smaller than 0.05.

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Problem 3 Chebyshev's inequality (6 credits)

a) State Chebyshev's inequality, including all required assumptions. No proof is required.

0 1 2 3

Let X be a real-valued random variable with finite mean $\mathbb{E}[X]$ and finite variance $\mathrm{var}[X] > 0$. Then for any $\epsilon > 0$,

$$P(|X - \mathbb{E}[X]| \geq \epsilon) \leq \frac{\mathrm{var}[X]}{\epsilon^2} .$$

0.5 points each for the finite mean, the finite positive variance, and $\epsilon > 0$; 1.5 points for the correct probability bound, including centering, inequality direction, and the squared threshold. Award partial credit within the 1.5 statement points as appropriate. Accept equivalent notation, the valid generalization allowing zero variance, the variant with $>$ instead of $\geq$, or a correctly stated standardized form. A finite-second-moment assumption covers both finiteness conditions. No density assumption is needed.

b)* For each sample size $n \geq 1$, an estimator $\hat{\theta}_n$ of a real parameter θ satisfies

$$\mathbb{E}_\theta[\hat{\theta}_n] = \theta, \qquad \mathrm{var}_\theta[\hat{\theta}_n] \leq \frac{4}{n}$$

for every $\theta \in \Theta$. Determine the smallest integer n for which the bound from Chebyshev's inequality guarantees

$$P_\theta\big(|\hat{\theta}_n - \theta| < 1/2\big) \geq 0.95 \qquad \text{for every } \theta \in \Theta .$$

0 1 2 3

For each θ, the estimator is centered at θ. Therefore,

$$P_\theta\big(|\hat{\theta}_n - \theta| \geq 1/2\big) \leq \frac{\mathrm{var}_\theta[\hat{\theta}_n]}{(1/2)^2} \leq \frac{16}{n} .$$

Taking complements gives $P_\theta(|\hat{\theta}_n - \theta| < 1/2) \geq 1 - 16/n$. The desired guarantee follows whenever

$$\frac{16}{n} \leq 0.05 = \frac{1}{20}, \qquad \text{i.e.,} \qquad n \geq 320 .$$

Thus the smallest integer permitted by this bound is $n = 320$.
Equivalently, unbiasedness gives $\mathbb{E}_\theta[(\hat{\theta}_n - \theta)^2] = \mathrm{var}_\theta[\hat{\theta}_n] \leq 4/n$, so Markov's inequality at threshold $1/4$ yields the same bound $16/n$.
1 point for correctly applying Chebyshev to the centered estimator at threshold $1/2$; 0.5 points for the bound $16/n$; 0.5 points for requiring $16/n \leq 0.05$ (or the equivalent coverage inequality); 1 point for solving and concluding that the smallest integer is $n = 320$. Accept a complete direct argument using Markov's inequality on the squared error with the same allocation. Award subsequent method points after a single arithmetic error, provided the approach is otherwise valid. A normal approximation does not give the requested finite-sample guarantee. No conclusion about an actually necessary sample size for a particular distribution is required.

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Problem 4 Server modes (8 credits)

During a fixed time interval, a server operates in a single mode $M \in \{0,1\}$, with $P(M = 1) = q$ and $P(M = 0) = 1 - q$, where $q \in (0,1)$. Conditionally on $M = 0$, the number of requests X follows $\mathrm{Poi}(\lambda_0)$; conditionally on $M = 1$, it follows $\mathrm{Poi}(\lambda_1)$, where $0 < \lambda_0 < \lambda_1$. Recall that for $Z \sim \mathrm{Poi}(\lambda)$, $P(Z = k) = e^{-\lambda}\lambda^k/k!$ for $k \in \mathbb{N}$, and $\mathbb{E}[Z] = \mathrm{var}[Z] = \lambda$.

0 1

a) Compute $\mathbb{E}[X]$ in terms of q, λ_0, and λ_1.

The law of total expectation gives

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid M]] = (1 - q)\lambda_0 + q\lambda_1 .$$

0.5 points for weighting the two conditional means by their mode probabilities; 0.5 points for the correct final expression. An immediately stated correct weighted sum receives the full point. A direct computation from the marginal pmf is equally valid.

0 1 2

b)* Compute $P(M = 1 \mid X = 0)$ in terms of q, λ_0, and λ_1.

By Bayes' rule and the law of total probability,

$$P(M = 1 \mid X = 0) = \frac{P(X = 0 \mid M = 1)P(M = 1)}{P(X = 0)} = \frac{qe^{-\lambda_1}}{(1 - q)e^{-\lambda_0} + qe^{-\lambda_1}} .$$

The denominator is strictly positive.
1 point for the correct Bayes numerator $qe^{-\lambda_1}$; 1 point for the total-probability denominator and the correct ratio. A correct unsimplified expression in Poisson probabilities receives full credit. Merely quoting Bayes' rule without identifying the events earns 0.5 points.

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c)* Compute var[X] in terms of q, λ_0, and λ_1.

0 1 2 3

For a Poisson random variable, $\mathbb{E}[Z^2] = \text{var}[Z] + \mathbb{E}[Z]^2 = \lambda + \lambda^2$. Splitting by the two modes therefore gives

$$\mathbb{E}[X^2] = (1-q)(\lambda_0 + \lambda_0^2) + q(\lambda_1 + \lambda_1^2)\,.$$

Writing $\mu := (1-q)\lambda_0 + q\lambda_1$, we obtain

$$\begin{aligned}\text{var}[X] &= \mathbb{E}[X^2] - \mu^2 \\ &= \mu + (1-q)\lambda_0^2 + q\lambda_1^2 - \big((1-q)\lambda_0 + q\lambda_1\big)^2 \\ &= \mu + q(1-q)\big(\lambda_0^2 - 2\lambda_0\lambda_1 + \lambda_1^2\big) \\ &= (1-q)\lambda_0 + q\lambda_1 + q(1-q)(\lambda_1 - \lambda_0)^2\,.\end{aligned}$$

Alternatively, the law of total variance gives the expected conditional variance $(1-q)\lambda_0 + q\lambda_1$ plus the variance of the conditional mean, $\text{var}[\lambda_0 + (\lambda_1 - \lambda_0)M] = q(1-q)(\lambda_1 - \lambda_0)^2$.

1 point for the correct second moment; 1 point for subtracting the squared mean with the correct cross term; 1 point for the resulting variance (an equivalent unsimplified expression is sufficient). For a total-variance argument: 1 point for the expected conditional variance; 1.5 points for the variance of the conditional mean; 0.5 points for combining the terms. If an error from (a) is consistently carried forward, deduct for that error only once; remaining method credit is available. A probability mixture is not the deterministic weighted sum $(1-q)Z_0 + qZ_1$ of independent Poisson variables.

d) Can the marginal distribution of X be a Poisson distribution? Justify your answer.

0 1 2

No. Every Poisson distribution has equal mean and variance. Here $q(1-q)(\lambda_1 - \lambda_0)^2 > 0$ by the assumptions, so $\text{var}[X] > \mathbb{E}[X]$. Thus X cannot have a Poisson distribution with any rate.

1 point for the equality of mean and variance in a Poisson distribution; 1 point for the strict inequality here and the conclusion. A bare assertion that a mixture is never Poisson is insufficient: the strict assumptions on q and the two rates matter. Accept any other valid proof. If an earlier erroneous expression is used consistently, award valid follow-through reasoning without a repeated deduction; do not award a new conceptual error.

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Problem 5 **Absolute measurement errors (13 credits)**

We are testing a voltmeter by shorting its two input terminals, so the true voltage is zero. Due to measurement errors, the device still measures small random positive and negative values. We model these measurements as $X_1, \dots, X_n$ iid $\mathcal{N}(0, \theta)$ with unknown parameter $\theta > 0$. However, in our test software, we actually only store the magnitudes $Y_i := |X_i|$ of the measurements.

0 1 2 3

a) Derive a probability density function $p_\theta(y)$ of Y_1, including its support.

For $y < 0$, $P(Y_1 \le y) = 0$. Write F_θ and f_θ for the cdf and density of X_1. For $y \ge 0$,

$$P(Y_1 \le y) = P(-y \le X_1 \le y) = F_\theta(y) - F_\theta(-y)\,.$$

Thus for $y > 0$, differentiation and the symmetry of f_θ yield $p_\theta(y) = f_\theta(y) + f_\theta(-y) = 2f_\theta(y)$. A density is therefore

$$p_\theta(y) = \begin{cases} \sqrt{\dfrac{2}{\pi\theta}}\exp\left(-\dfrac{y^2}{2\theta}\right), & y \ge 0, \\ 0, & y < 0\,. \end{cases}$$

0.5 points for the correct support; 1 point for the event or cdf calculation; 1 point for correctly differentiating both terms, including the factor 2; 0.5 points for the final density. Accept an equivalent derivation by summing the two inverse branches or a justified symmetry argument. Any value at $y = 0$ is acceptable.

0 1 2 3

b)* For an observed sample $y_1, \dots, y_n$, derive the maximum likelihood estimator of θ and justify that it maximizes the likelihood over $\theta > 0$.
Hint: If you could not solve part (a), you may assume that $Y_1, \dots, Y_n$ are iid with the alternative density

$$\widetilde{p}_\theta(y) = c\,\theta^{-1/4}\exp\left(-\frac{y^4}{2\theta}\right) \quad \text{for } y \ge 0, \qquad \widetilde{p}_\theta(y) = 0 \quad \text{for } y < 0\,,$$

where $c > 0$ is a normalizing constant independent of θ. Use this alternative model only in part (b).

Original model. Let $S_2 := \sum_{i=1}^n y_i^2 > 0$.

$$L(\theta \mid y) = \left(\frac{2}{\pi\theta}\right)^{n/2} e^{-S_2/(2\theta)},$$

$$\ell(\theta \mid y) = C - \frac{n}{2}\ln\theta - \frac{S_2}{2\theta}, \qquad \ell'(\theta \mid y) = \frac{S_2 - n\theta}{2\theta^2}\,.$$

The derivative changes from positive to negative at S_2/n, so $\widehat{\theta}_{\text{mle}} = n^{-1}\sum_{i=1}^n Y_i^2$ is the unique global maximizer.
Alternative model. Let $S_4 := \sum_{i=1}^n y_i^4 > 0$.

$$\widetilde{L}(\theta \mid y) = c^n\theta^{-n/4}e^{-S_4/(2\theta)},$$

$$\widetilde{\ell}(\theta \mid y) = \widetilde{C} - \frac{n}{4}\ln\theta - \frac{S_4}{2\theta}, \qquad \widetilde{\ell}'(\theta \mid y) = \frac{2S_4 - n\theta}{4\theta^2}\,.$$

The derivative changes from positive to negative at $2S_4/n$, so $\widehat{\theta}_{\text{alt}} = 2n^{-1}\sum_{i=1}^n Y_i^4$ is the unique global maximizer. Here C and $\widetilde{C}$ are independent of θ. Either model earns full credit; only one is required. Award 0.5 points each for the iid likelihood, log-likelihood, derivative, likelihood equation solution, justification of the maximum, and final estimator. Accept direct maximization, a negative second derivative at the unique stationary point, or concavity after reparameterization. Constants independent of θ may be dropped; don't deduct points for follow on errors from (a).

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c)* In the original model (regardless of what you used in (b)), is the maximum likelihood estimator of θ unbiased? Justify your answer and compute its variance.
For this subproblem, you may use $\mathbb{E}[Y_i^2] = \theta$ and $\mathbb{E}[Y_i^4] = 3\theta^2$.
Hint: If you could not solve part (b) for the original model, you may continue with $\widehat{\theta}_{\text{mle}} = n^{-1}\sum_{i=1}^{n} Y_i^2$.

0 1 2

By linearity of expectation,

$$\mathbb{E}[\widehat{\theta}] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[Y_i^2] = \theta\,,$$

so $\widehat{\theta}$ is unbiased. Since $\operatorname{var}[Y_i^2] = \mathbb{E}[Y_i^4] - (\mathbb{E}[Y_i^2])^2 = 2\theta^2$ and the Y_i^2 are independent,

$$\operatorname{var}[\widehat{\theta}] = \frac{1}{n^2}\sum_{i=1}^{n}\operatorname{var}[Y_i^2] = \frac{2\theta^2}{n}\,.$$

0.5 points for the expectation calculation; 0.5 points for the unbiasedness conclusion; 0.5 points for $\operatorname{var}[Y_i^2] = 2\theta^2$; 0.5 points for the variance of the sample mean, using independence. Accept equivalent direct second-moment calculations. Do not deduct twice for one error. Students who used the alternative density in (b) should use the continuation estimator here, under the original Gaussian model. For an erroneous original-model estimator from (b), award correct subsequent calculations and conclusions without repeating the earlier deduction.

d)* An engineer wants to determine whether the variance θ of the *signed* errors exceeds 1. For $n = 200$ independent measurements, the observed *mean squared absolute error* is $\frac{1}{200}\sum_{i=1}^{200} y_i^2 = \frac{6}{5}$. Formulate the hypotheses and perform an approximate test at significance level $\alpha = 0.05$ using the central limit theorem and the variance at the null boundary. State the test statistic, its approximate distribution at the null boundary, the rejection rule, its observed value, and your conclusion.
For this subproblem, you may use that $W_i := Y_i^2$ are iid with $\mathbb{E}[W_i] = \theta$ and $\operatorname{var}[W_i] = 2\theta^2$. The 0.95-quantile of $\mathcal{N}(0,1)$ is $z_{0.05} \approx 1.64$.

0 1 2 3 4 5

The hypotheses are $H_0 : 0 < \theta \le 1$ against $H_1 : \theta > 1$. Let $\overline{W}_{200} := 200^{-1}\sum_{i=1}^{200} W_i$. At the null boundary $\theta_0 = 1$,

$$\mathbb{E}[\overline{W}_{200}] = 1, \qquad \operatorname{var}[\overline{W}_{200}] = \frac{2}{200} = \frac{1}{100}\,.$$

By the clt, the statistic

$$Z := \frac{\overline{W}_{200} - 1}{\sqrt{2/200}} = 10(\overline{W}_{200} - 1)$$

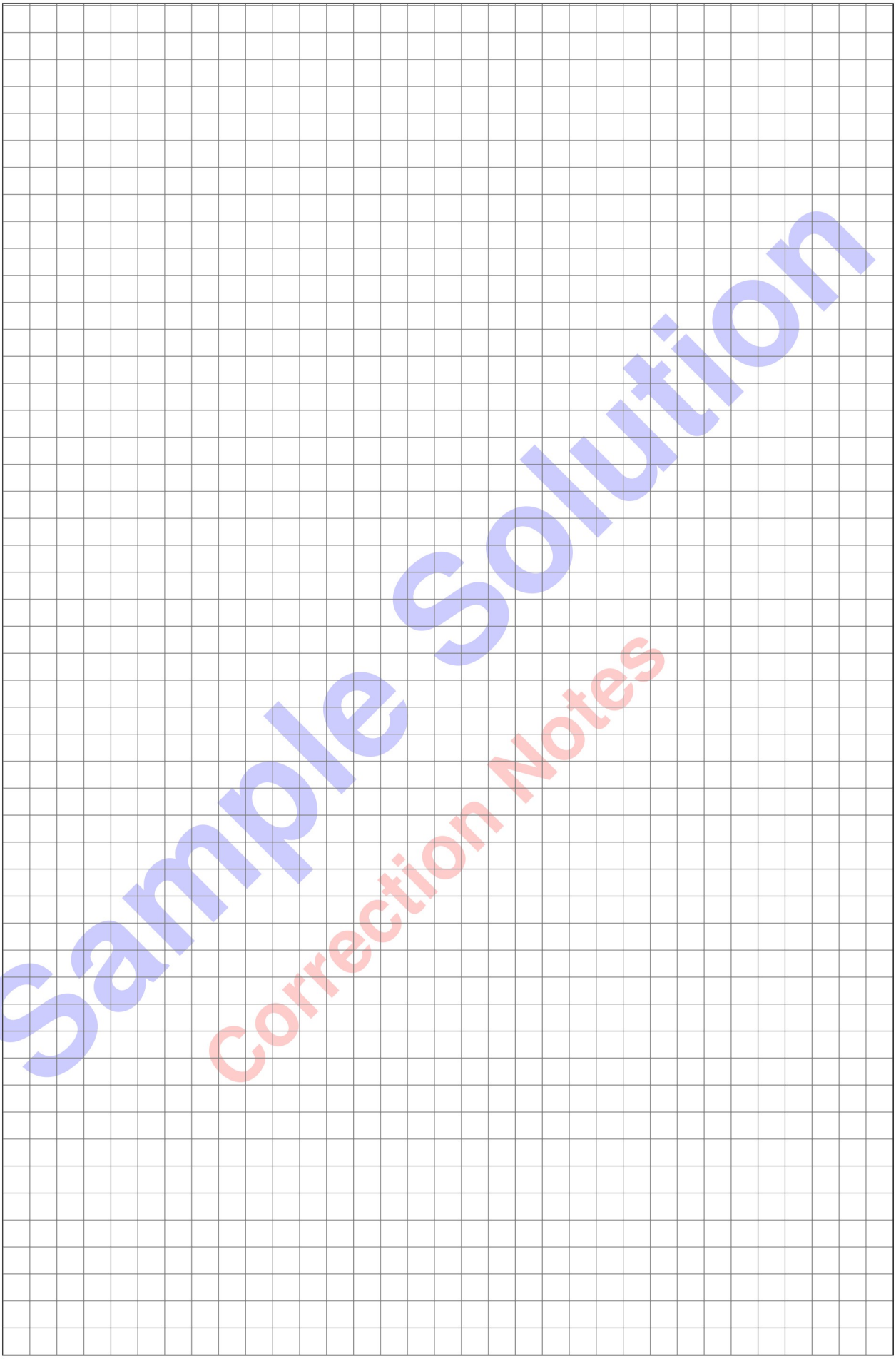
is approximately $\mathcal{N}(0,1)$ at $\theta = 1$. We reject H_0 if $z_{\text{obs}} > z_{0.05} \approx 1.64$, equivalently if $\overline{w}_{200} > 1.164$. For the observed sample,

$$z_{\text{obs}} = 10\left(\frac{6}{5} - 1\right) = 2 > 1.64\,.$$

We reject H_0 at the approximate significance level 0.05. The data provide evidence that the variance of the signed errors exceeds 1.
The boundary $\theta = 1$ is the least favorable point of the null: writing $X_i = \sqrt{\theta}\,G_i$ with iid $G_i \sim \mathcal{N}(0,1)$ gives $\overline{W}_{200} = \theta\,200^{-1}\sum_i G_i^2$. Its probability of exceeding a fixed positive threshold increases with θ.
0.5 points each for the null and alternative; 0.5 points each for the null-boundary mean and variance (or standard error); 0.5 points each for the standardized statistic and its approximate standard Gaussian distribution at $\theta = 1$; 0.5 points each for the upper-tail rejection rule and observed value 2; 0.5 points each for the decision and its interpretation. Total: 5 points. A proof that the boundary is least favorable is not required. Accept an equivalent unstandardized rejection rule or an exact chi-squared test with a justified critical comparison and decision. A claim that the Gaussian approximation is exact loses only its 0.5 distribution point. After one arithmetic or tail-direction error, award correct subsequent method points without repeated deductions. Claiming that H_1 has probability 0.95 is not a correct interpretation.

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Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.



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